Bonus Project: Deep Learning (SS26)

Multivariate Time Series Forecasting

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Overview

This semester's bonus project focuses on **multivariate time series forecasting**. You will be given a benchmark dataset containing multiple correlated time series. Your task is to develop a deep learning model that produces accurate forecasts, beat our provided baseline, and write a concise report about your approach.

Task Description

1. **Model Development.** Design and implement a deep learning model for multivariate time series forecasting. You are free to choose any architecture (e.g., Transformers, state-space models, temporal convolutional networks, recurrent networks, or hybrid approaches) and any technique (e.g., pre-training, data augmentation, ensembling). The model should be implemented in PyTorch.

2. **Benchmark Evaluation.** We will provide a multivariate time series benchmark dataset along with a predefined train/validation/test split. Details on the dataset (number of variates, history length, rollout horizon, required prediction horizon, evaluation metric, dataset link, and file schema) will be announced on Moodle.

   Your prediction file must contain one prediction for every row in the provided forecast index. If the released rollout horizon is shorter than the full forecast index, you must roll your model forward until all required timestamps are predicted. We provide a *naive last-value* baseline as the minimum reference to beat and stronger seasonal and lag baselines for comparison. Your model must improve upon the naive baseline on the held-out test set.

3. **Additional Dataset Experiment.** To test whether your modeling choices generalize beyond the benchmark, each group must propose *one* further time series dataset that could be used for forecasting, e.g., from finance, industry, energy, healthcare, traffic, weather, or another domain.

   Briefly describe the dataset, its variables, the forecasting use case, and what makes it suitable and interesting for a deep learning forecasting task. You must run your architecture on this additional dataset and report the result. The additional dataset does not need to use exactly the same preprocessing, horizon, or metric as the benchmark dataset, but your experimental setup must be clearly described. You are encouraged to compare against further model variants, baselines, or ablations if this helps support your analysis.

4. **Validation Leaderboard.** We will host an automatic leaderboard on Hugging Face Spaces. You may submit validation predictions to receive immediate feedback and see your group on the public validation leaderboard. The private test set will not be distributed.

5. **Final Private Evaluation.** For the final evaluation, you will submit a model archive rather than test predictions. Your archive must contain a reproducible PyTorch inference script, dependencies, and checkpoint files. We will run your submitted model on the private test inputs and compute the final score ourselves. The required inference command is:

```
python predict.py --input_dir /data/input --output_file
/output/predictions.csv --checkpoint /submission/checkpoint.pt
```

6. **Exposé.** On 8. June 2026, each group must submit a one-page exposé outlining their planned project direction. It should briefly describe the additional dataset, the intended model architecture or model family, planned improvements or variants, baselines for comparison, key risks or open questions, and the main sources to be used. Treat the exposé as an early draft of the final report.

7. **Report.** Write a report (4–6 pages, excluding references) in LaTeX that covers:

   - **Introduction:** What problem are you solving and why is it interesting?
   - **Related Work:** What existing approaches are relevant? Cite and briefly discuss them. This section should build on the sources identified in your exposé.
   - **Method:** Describe your model architecture, input features, training setup, and any design decisions. Include a clear figure or diagram of the architecture. Explain how your final method differs from, or follows, the plan in your exposé.
   - **Experiments on the benchmark dataset:** Present your validation results on the provided benchmark dataset **and** your choosen dataset. Compare against the naive last-value baseline, the seasonal mean baseline, and any other methods you tried. Include ablation studies or analyses that give insight into why your approach works (or does not).
   - **Additional Dataset:** Describe your proposed additional dataset and report how your architecture performs on it. Discuss whether the model transfers well, what had to be changed, and what this reveals about the method.
   - **Conclusion:** Summarize your findings and discuss limitations or future directions.

   Please use clean and complete references to the literature.

8. **Code.** Submit your code as a well-structured repository (e.g., a zip file or a link to a Git repository). Include a `README.md` with instructions on how to reproduce

your results. Your final model archive must follow the submission template released with the project materials.

Deliverables

Each group must submit:

1. A one-page **exposé** as a PDF.
2. The final **report** as a PDF.
3. The **code** with a README for reproducibility.
4. The **final model archive** via the leaderboard submission system.

Submission System

The leaderboard and submission system will be hosted on Hugging Face Spaces at `https://huggingface.co/spaces/AIML-TUDA/dlam-ts-project-leaderboard-2026`. Each group must log in with Hugging Face before submitting. Validation submissions are scored automatically against hidden validation labels. Final model archives are stored by the submission system and evaluated privately by the instructors after the deadline. For the Report you will upload a PDF file to Moodle. The prediction CSV format, metric, and dataset link will be announced on Moodle.

Groups

The project should be done in groups of **2 to 4 students**. The project should be large enough so that every member can contribute meaningfully. Clearly state each member's contribution in the report.

Bonus

The project can grant a **bonus between 0.3 and 0.7 on your final exam grade**. The exact bonus is determined based on the overall quality of the submission. To be eligible for any bonus, the following minimum requirements must be met:

- Your model beats the naive last-value baseline on the private test set.
- Your report is of sufficient quality, with clear writing, meaningful experiments, and complete references.
- Your code and final model archive are reproducible and follow the provided submission requirements.

Submissions that substantially exceed the minimum requirements, for example through strong empirical performance, careful analysis, well-motivated modeling choices, convincing additional experiments, or particularly clear reporting, may receive a higher bonus.